

Can Simple SDEs Describe Equity-Market Time Series?

A Mini-Research with Diffusion and Ornstein–Uhlenbeck Benchmarks

Dmitri Bolt

Educational Mini-Research

May 2026

Research Question and Financial Setup

Central question

Can stock-market time series be described by simple textbook SDEs?

Financial setup (*for the SDE-oriented reader*):

- A **stock** is a tradeable financial asset (share of a listed company); its **closing price** $P_t^k > 0$ is observed once per **trading day** (≈ 252 days/year).
- Dataset: 94 large-cap US stocks, NASDAQ, Jan 2023 – Apr 2026.

Key insight: the same price series yields five structurally different objects:

$$P_t^k \rightarrow X_t^k = \ln P_t^k \rightarrow \Delta X_{t_i}^k \rightarrow \tilde{X}_t^k \rightarrow \Delta \tilde{X}_{t_i}^k$$

A benchmark SDE may be useful for one transformation and inadequate for another.

Three Textbook Benchmarks (Øksendal)

Following Øksendal, *Stochastic Differential Equations*, 6th ed.

① Itô diffusion

Def. 4.1.1, eq. (4.1.6)

$$dX_t = u dt + v dB_t$$

Benchmark for daily log-returns $\Delta X_{t_i}^k$.

② Geometric Brownian Motion

Ch. 5.1, eq. (5.1.5)

$$N_t = N_0 \exp\left((r - \frac{1}{2}\alpha^2)t + \alpha B_t\right)$$

N_t = price level. Benchmark for raw log-prices X_t^k .

③ Ornstein–Uhlenbeck (Exercise 5.7)

$$dX_t = \theta(m - X_t) dt + \sigma dB_t$$

$\theta > 0$: mean-reversion speed, m : equilibrium. Benchmark for detrended log-prices $\tilde{X}_t^k$.

Dataset

- 94 large-cap US stocks, daily prices
- NASDAQ, Jan 2023 – Apr 2026 (≈ 820 obs.)
- 3 representative stocks: **NVDA**, **META**, **GOOG**

Detrending: OLS linear fit to X_t^k , keep residual:

$$\tilde{X}_t^k = X_t^k - (\hat{a} + \hat{b}t)$$

Different transformations $\rightarrow$ different benchmarks.

Five-stage pipeline

P_t^k	prices
$\downarrow \ln$	
X_t^k	log-prices [<i>GBM</i>]
$\downarrow \Delta$	
$\Delta X_{t_i}^k$	log-returns [<i>Itô</i>]
$\downarrow$ OLS detrend	
$\tilde{X}_t^k$	detrended levels [<i>OU</i>]
$\downarrow \Delta$	
$\Delta \tilde{X}_{t_i}^k$	detrended returns

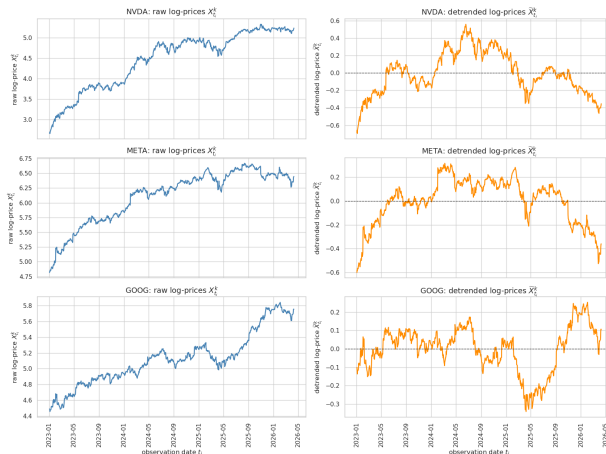
Representative Stocks: Screening

Selection criteria:

- High raw-trend R^2 : log-price well-described by linear growth $\Rightarrow$ GBM relevant
- High lag-1 ACF of $\tilde{X}^k$: high persistence $\Rightarrow$ OU slow mean reversion
- Large detrended amplitude: visually clear contrast

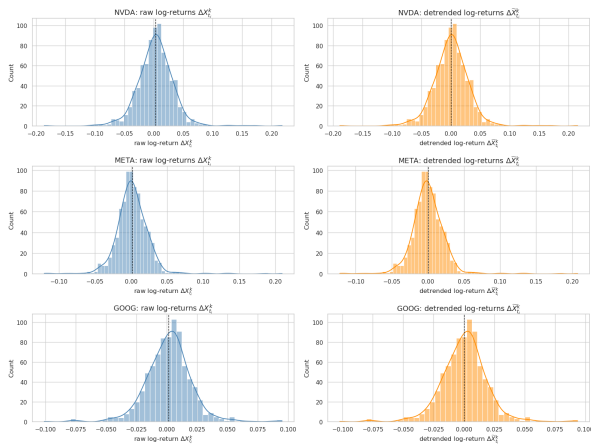
Sym.	Raw R^2	ACF ₁	Detr. std	Zer. cr.	Role
NVDA	0.883	0.9916	0.236	24	Strongest trend, large ampl.
META	0.822	0.9919	0.187	28	Growth, max persistence
GOOG	0.866	0.9880	0.121	41	Calmer, expressive
AAPL	0.782	0.9817	0.084	44	Backup (weaker)

Raw Log-Prices vs Detrended Log-Prices



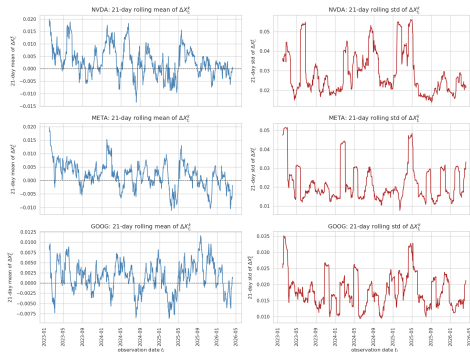
Left: raw $X_{t_i}^k$ — drift-dominated $\Rightarrow$ GBM benchmark. **Right:** detrended $\tilde{X}_{t_i}^k$ — oscillates near zero $\Rightarrow$ OU benchmark.

Increment Distributions: Daily Log>Returns



Left: raw $\Delta X_{t_i}^k$. **Right:** detrended $\Delta \tilde{X}_{t_i}^k$. Detrending shifts the *mean* toward zero but does not reshape the noise distribution.

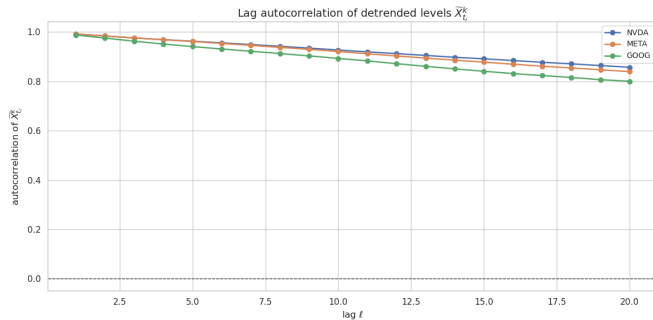
Rolling Diagnostics: Is the Diffusion Constant?



21-day rolling mean (left) and rolling std (right) of $\Delta X_{t_i}^k$:

- Rolling mean ≈ 0 ✓ — drift hard to detect (consistent with Itô)
- Rolling std is **time-varying** × — violates constant- σ : NVDA $\times 3.7$, META $\times 6.8$, GOOG $\times 3.7$

OU Persistence: ACF of Detrended Levels



OU theory (Ex. 5.7) predicts:

$$\text{corr}(\tilde{X}_t, \tilde{X}_{t-\ell}) = e^{-\theta\ell}$$

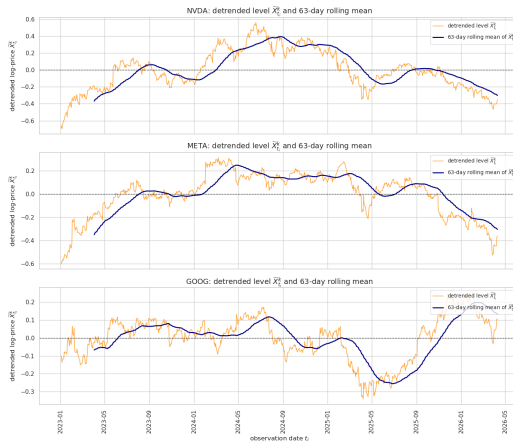
geometric decay in lag ℓ .

What the data show:

- High lag-1 ACF:
NVDA: 0.9916
META: 0.9919
GOOG: 0.9880
- Gradual step-by-step decay

$$\hat{\theta} \approx 0.012, \quad t_{1/2} \approx 57 \text{ days.}$$

Detrended Levels and Rolling Mean



Orange: $\tilde{X}_{t_i}^k$. Navy: 63-day rolling mean. Dashed: zero. The rolling mean itself drifts slowly — very slow mean reversion, not a fixed equilibrium.

OU Parameter Estimation: AR(1) Proxy

Bridge from continuous OU to discrete daily data:

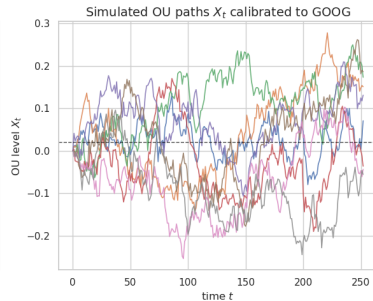
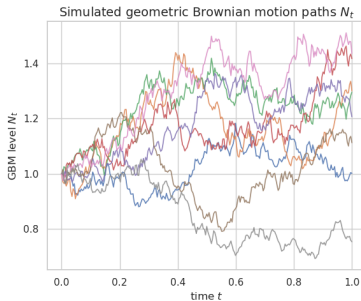
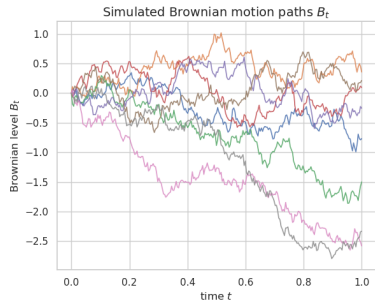
$$dX_t = \theta(m - X_t) dt + \sigma dB_t \xrightarrow{\text{one step}} \tilde{X}_{t+1} \approx \hat{\alpha} + \hat{\phi} \tilde{X}_t + \varepsilon_t$$

$$\hat{\phi} = e^{-\hat{\theta}}, \quad \hat{m} = \hat{\alpha}/(1 - \hat{\phi}), \quad t_{1/2} = \ln 2/\hat{\theta}$$

Symbol	$\hat{\phi}$	$\hat{\theta}$	$t_{1/2}$ (days)	$\hat{m}$	$\hat{\sigma}$
NVDA	0.9878	0.0123	56.4	0.0337	0.0305
META	0.9879	0.0122	56.8	0.0245	0.0236
GOOG	0.9880	0.0121	57.4	0.0207	0.0187

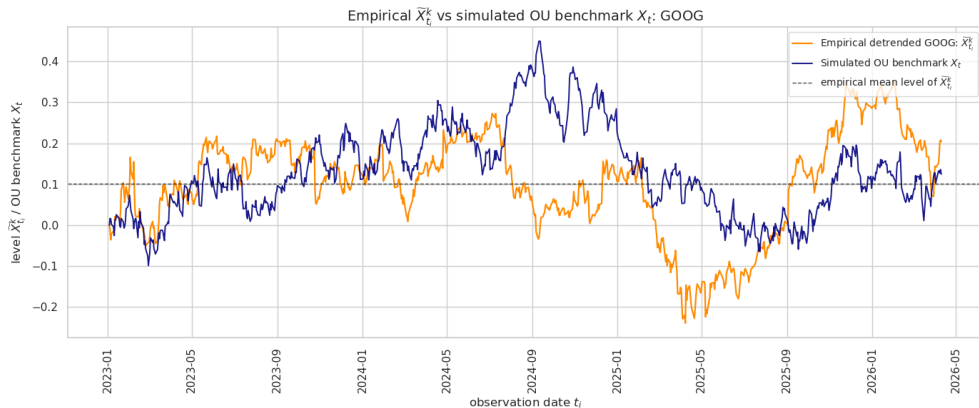
- $\hat{\phi} < 1$: valid slow mean reversion (not unit root)
- $t_{1/2} \approx 57$ trading days $\approx$ **3 calendar months**: gentle, gradual drift

Simulations: Connecting Theory and Data



L: B_t — pure noise. **C:** GBM N_t — multiplicative growth under noise. **R:** OU X_t calibrated to GOOG ($\hat{\theta} = 0.012$, $\hat{m} = 0.021$, $\hat{\sigma} = 0.019$).

Empirical GOOG vs Simulated OU Benchmark



Orange: empirical detrended GOOG $\tilde{X}_{t_i}^k$. Navy: one simulated OU path with fitted parameters. Visual comparison, not a statistical test.

What does “compatible” mean?

An explicit criterion is needed — visual similarity alone is not sufficient.

Assessment workflow:

- ① **Fit** using one simple statistic:
 - One-step AR(1) on detrended levels $\Rightarrow \hat{\alpha}, \hat{\phi}$
- ② **Validate** against implications **not used** in the fit:
 - ACF decay: empirical ACF at lags 2–20 vs. predicted $\hat{\phi}^\ell$
 - Transition mean: $\mathbb{E}[\tilde{X}_{t+h} \mid \tilde{X}_t]$ at $h = 5$ vs. model prediction $\hat{m} + \hat{\phi}^h(\tilde{X}_t - \hat{m})$
- ③ **Report the known failure:** time-varying volatility (rolling-vol ratio $> 3\times$) lies outside the model.

Model Assessment: Quantitative Summary

Sym.	<i>Fit</i>		$t_{1/2}$	<i>Validation</i>		<i>Failure</i>
	Raw R^2	OU R^2		ACF RMSE	Trans. RMSE	Vol. ratio
NVDA	0.883	0.983	56.4 d	0.049	0.0055	4.1×
META	0.822	0.984	56.8 d	0.038	0.0052	6.8×
GOOG	0.866	0.976	57.4 d	0.007	0.0069	3.7×

- **Raw R^2 high** — GBM benchmark applies to log-prices.
- **OU $R^2 \approx 0.98$** — one-step AR(1) fit is very tight.
- **ACF RMSE small** (GOOG: 0.007) — $\hat{\phi}^\ell$ tracks empirical decay. **Validation 1 passes.**
- **Trans. RMSE ≈ 0.005 – 0.007** — conditional mean accurate. **Validation 2 passes.**
- **Vol. ratio $> 3\times$** — constant- σ SDE fails here (expected).

Main Findings

The answer is qualified, not absolute:

- ① **GBM $\Rightarrow$ raw log-prices** ($R^2 \approx 0.82\text{--}0.88$)
Raw log-prices are trend-dominated; multiplicative-growth benchmark applies.
- ② **OU $\Rightarrow$ detrended log-prices** (lag-1 ACF $\approx 0.988\text{--}0.992$)
Half-life $t_{1/2} \approx 57$ days (≈ 3 months). ACF-decay and conditional-mean validations both pass.
- ③ **Itô diffusion $\Rightarrow$ increments, but σ is not constant**
Rolling-vol ratio $> 3\text{--}7\times$: volatility clustering is outside the model.

No single low-dimensional SDE simultaneously explains all features.

Limitations and Natural Extensions

Current limitations

- ① Educational diagnostic, not formal likelihood comparison
- ② Simple linear detrending (not adaptive, not factor-based)
- ③ Daily data only
- ④ Constant- σ misses *volatility clustering*
- ⑤ Transition-mean check is simplified; not a full Markov transition-kernel test

Natural next steps

- ① Realized quadratic variation: directly measures $\langle X \rangle_t$
- ② PCA market-factor extraction; OU on detrended factor
- ③ Stochastic-volatility extension (σ_t follows a diffusion)
- ④ Jump-diffusion models for large overnight moves
- ⑤ Fokker–Planck and full transition-kernel analysis
- ⑥ MLE-based parameter estimation and formal model comparison

Conclusion

Answer

Stock-market time series **can** be described by SDEs to a meaningful extent, but the description is *approximate* and depends on which object is modeled.

- Raw X_t^k (log-prices) $\rightarrow$ **GBM** is the appropriate benchmark.
- Detrended $\tilde{X}_t^k \rightarrow$ **OU** with $t_{1/2} \approx 57$ d.
- Increments $\Delta X_{t_i}^k \rightarrow$ **Itô diffusion** language is natural, but σ is time-varying.

Methodological lesson

“Compatible” = fit by one set of statistics, validate by others.

Data: NASDAQ daily prices, Jan 2023 – Apr 2026, 94 stocks. Theory: Øksendal, *SDE*, 6th ed.